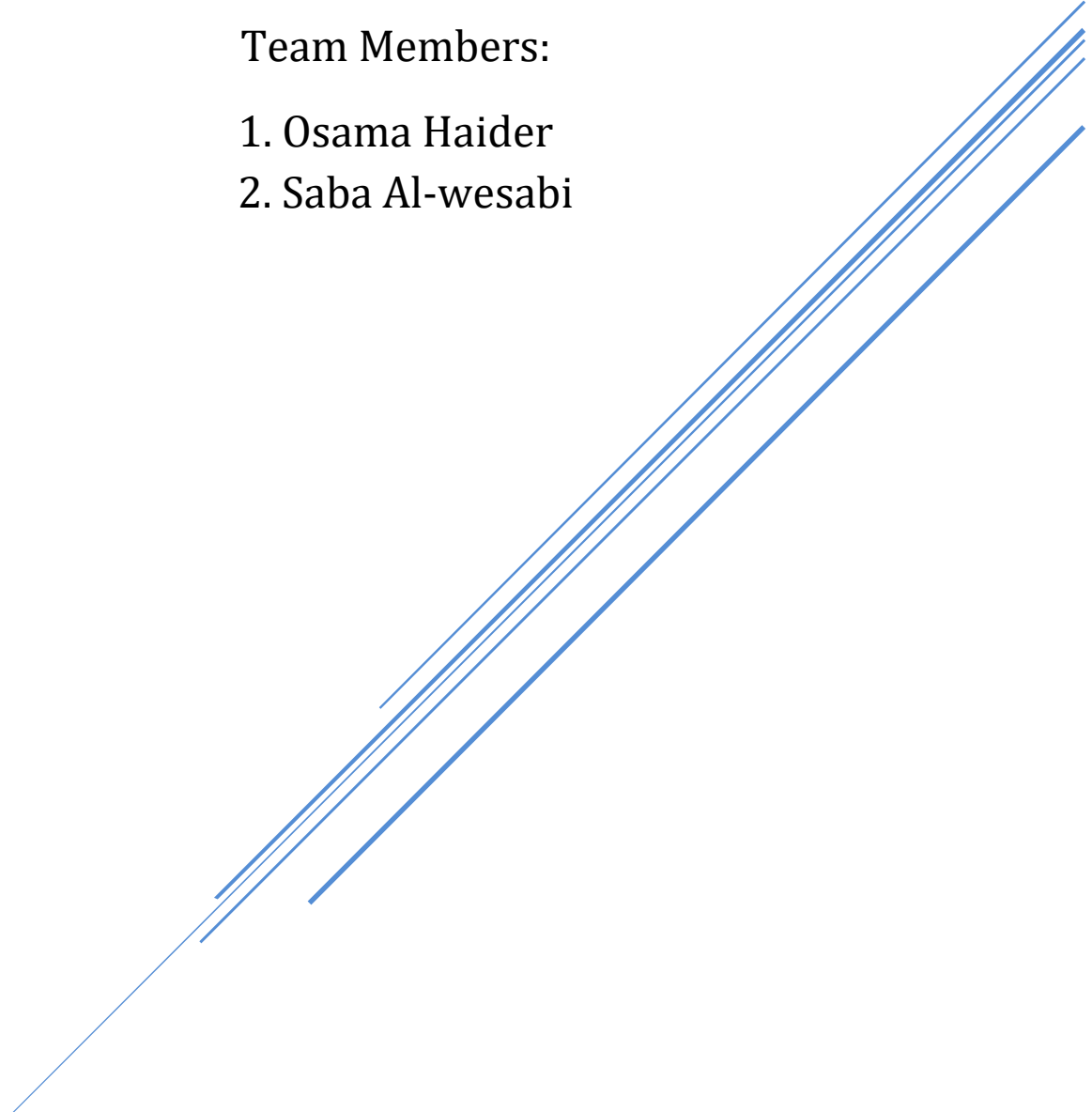


# SMART CAMPUS IOT SIMULATION AND ANALYTICS PLATFORM

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Internet of Things  
Master of IT

## Abstract

This Internet of Things project proposes a pure-simulation Internet of Things framework for smart campus management and control purposes. As opposed to a traditional IoT project implementation with real hardware boards and actual sensors, this solution uses Python-based virtual devices to produce simulated data readings from various campus rooms. The virtual devices publish the produced information through MQTT Broker using a particular topic hierarchy structure. The Python Subscriber module gets data messages, verifies the data payload, calculates analytics values, and generates alerts for out-of-normal situations. Node-RED Dashboard flow illustrates real-time temperatures, humidity levels, occupancy rate, alerts, and temperature trend.

The chosen wireless communication technology and networking model are specified as IEEE 802.15.4 / Zigbee-like protocols, which are very well suited to indoor university campus sensor networks with small-periodic-packets at low power consumption rates. In order to achieve maximum simulation realism of the process, a software-based simulator includes packet loss rates, random propagation and queueing delays, RSSI and LQI metrics, dynamic cumulative energy consumption (mWh), and battery life depletion curves. This academic report discusses the design approaches, protocol architecture, verification procedures, and visualization layers in detail.

## 1. Problem Statement and Scenario

In today's campus environments, there is a complex array of spaces in the form of classrooms, lecture theaters, offices, scientific laboratories, and server rooms. All of these areas have unique environment and safety requirements as well as requirements for energy efficiency that need constant monitoring. For instance, lecture theaters have to have proper microclimates for increased levels of concentration of the students, the laboratory needs to have proper AQI control due to the presence of toxic chemicals, offices should have energy-efficient lighting based on occupancy control, and server rooms need microsecond temperature warning systems.

The use of physical prototypes as a form of testing may prove to be cost-prohibitive, highly complicated, and quite tedious in an academic context. This is why this particular project concentrates on a fully-fledged software-based IoT simulation framework which functions exactly the same way as the actual implementation would. This simulation represents a smart campus ecosystem with seven unique physical locations:

| Device ID   | Label       | Building             | Monitoring Purpose               |
|-------------|-------------|----------------------|----------------------------------|
| hall_1_1    | Hall 1.1    | floor_1              | Teaching – comfort & occupancy   |
| hall_1_2    | Hall 1.2    | floor_1              | Teaching – comfort & occupancy   |
| hall_1_3    | Hall 1.3    | floor_1              | Teaching – comfort & occupancy   |
| hall_1_4    | Hall 1.4    | floor_1              | Teaching – comfort & occupancy   |
| server_room | Server Room | it_building          | High-temperature risk monitoring |
| lab1        | Lab 1       | engineering_building | Lab comfort & air quality        |
| office1     | Office 1    | admin_building       | Energy saving & occupancy        |

## 2. IoT Architecture

In order to achieve modularity, scalability, and maintainability, the architecture adopted by the platform is the classical and well-known four-tier architecture of Internet of Things. This layer-based architecture replicates industrial architectures, like AWS IoT Core and Microsoft Azure IoT Solutions:

- **Perception Layer:** The Python virtual devices replicate the sensors for the measurement of the temperature, humidity, occupancy, light, air quality, battery status, and the device status.
- **Network Layer:** MQTT communication protocol is used to transfer the telemetry messages in a hierarchical topic structure through the HiveMQ Public Broker. The communication model resembles that of IEEE 802.15.4 / Zigbee communication standard.
- **Processing Layer:** The processing/subscriber.py file subscribes to messages with a wildcard topic, checks and processes each message, computes the derived values and assesses the alerts.
- **Application Layer:** A Node-RED Dashboard shows the current gauges, the alert text, and a graph of the temperature for all devices.

## 3. Wireless Communication Choice

The perception layer is based on IEEE 802.15.4 features on the physical and MAC layers represented in a simulation with a similar network profile to Zigbee. In contrast to other high-throughput LAN systems (Wi-Fi based on IEEE 802.11 standards) or wide-area outdoor approaches (LoRaWAN), Zigbee has been designed especially for low range, power consumption and data rate of mesh networks for building automation.

In contrast to high-bandwidth technologies such as Wi-Fi that would drain the batteries rapidly, the sensors installed in the room need only to transmit telemetry information without requiring a high bandwidth connection. LoRaWAN uses frequencies below 1 GHz (e.g., 868/915 MHz) with a focus on long-range communication up to several kilometers. It is thus over-designed, and legally constrained regarding duty cycle for frequent indoor transmission (every 5 seconds) due to its sub-GHz frequencies. Therefore, Zigbee channel access protocol (CSMA/CA), routing and sleep modes perfectly suit this project. Random delays, LQI/RSSI variations, packet losses and battery depletion have been incorporated into the python simulator to imitate the physical layer realistically.

### Comparison with Other Wireless Types

| Feature                 | Zigbee-like (Selected) | Wi-Fi (IEEE 802.11)  | LoRaWAN (LoRa Alliance) |
|-------------------------|------------------------|----------------------|-------------------------|
| <b>Frequency Band</b>   | 2.4 GHz ISM Band       | 2.4 / 5 / 6 GHz      | Sub-GHz (868/915 MHz)   |
| <b>Indoor Range</b>     | 10 - 100 meters        | 30 - 50 meters       | 1 - 5 kilometers        |
| <b>Power Profile</b>    | Extremely Low          | High (Not Battery)   | Very Low                |
| <b>Data Rate Limit</b>  | 250 kbps               | Up to 10+ Gbps       | Under 50 kbps           |
| <b>Typical Topology</b> | Mesh / Star-Mesh       | Star (Access Point)  | Star-of-Stars (Gateway) |
| <b>Suitability</b>      | High (Optimal)         | Medium (Power-heavy) | Low (Duty Cycle Limits) |

## 4. MQTT Topic Design and Payload Format

Selection of Application Layer Messaging Protocol, MQTT is chosen as an application layer messaging protocol. Contrary to HTTP, which works according to a request-response scheme involving significant overhead due to the HTTP headers and costs associated with the establishment and tear-down of TCP connections, MQTT operates under a publish-subscribe paradigm. The latter makes it a perfect fit for resource-constrained devices. MQTT broker takes care of routing the messages between clients (devices), and thus one client publishes telemetry messages to particular topics and another client (subscriber or Node-Red) receives them asynchronously.

For organizing telemetry in a logical manner within the virtual campus, a well-defined hierarchy of topics is used. Wildcards can be used within the hierarchy, and therefore the subscriber can collect information from all buildings, rooms, and devices using only one topic structure:

Topic hierarchy:

```
campus/{building}/{room}/{deviceId}/telemetry
```

Example:

campus/floor\_1/hall\_1\_1/hall\_1\_1/telemetry

Example JSON payload:

```
{
  "deviceId": "hall_1_1",
  "label": "Hall 1.1",
  "building": "floor_1",
  "room": "hall_1_1",
  "timestamp": "2026-07-07T09:00:00+00:00",
  "temperature": 26.4,
  "humidity": 58.2,
  "occupancy": 22,
  "light_level": 70.5,
  "air_quality": 67.3,
  "battery_level": 94.8,
  "status": "OK",
  "wireless": {
    "model": "IEEE 802.15.4 / Zigbee-like",
    "packet_loss_rate": 0.05,
    "random_delay_ms": 180,
    "link_quality": 88,
    "rssi": -63,
    "estimated_energy_consumption_mwh": 0.036
  }
}
```

## 5. Processing Logic and Rules

The edge processing layer, realized in `processing/subscriber.py`, deals with turning raw sensor values into usable information. Instead of sending raw information straight to the database and display, the subscriber validates data locally, parses payload metadata, and calculates additional derived variables. In this way, the subscriber lessens the burden on servers downstream and ensures real-time notifications through alarms.

There are two types of additional derived variables calculated:

1. **Comfort Index:** It is determined with the help of the Humidex approximation formula:  $\text{Temperature} + (\text{Humidity} / 100 * 5)$ . This variable offers a more precise measurement of thermal comfort than the temperature.
2. **Air Quality State:** It is classified according to Indoor Air Quality (IAQ) states: 'GOOD' when  $\text{IAQ} \leq 80$ , 'MODERATE' when  $\text{IAQ} \leq 120$ , and 'POOR' when  $\text{IAQ} > 120$ . The last state enables facility management to turn on ventilation.

### Derived Values

| Derived Value           | Mathematical/Logic Description                          |
|-------------------------|---------------------------------------------------------|
| <b>comfort_index</b>    | $\text{Temperature} + (\text{Humidity} / 100 \times 5)$ |
| <b>occupancy_status</b> | OCCUPIED if Occupancy > 0, otherwise<br>EMPTY           |

|                           |                                             |
|---------------------------|---------------------------------------------|
| <b>high_temp_flag</b>     | True when Temperature > 35, otherwise False |
| <b>air_quality_status</b> | GOOD (<=80), MODERATE (81-120), POOR (>120) |

### Alert Rules

| Alert                            | Condition                          |
|----------------------------------|------------------------------------|
| <b>High temperature</b>          | temperature > 35                   |
| <b>Empty room with lights on</b> | occupancy = 0 AND light_level > 60 |
| <b>Poor air quality</b>          | air_quality > 120                  |
| <b>Abnormal humidity</b>         | humidity > 80                      |
| <b>Low battery</b>               | battery_level < 20                 |

### Excel Persistent Data Storage (openpyxl Integration)

To ensure long-term data persistence without requiring heavy database installations, the subscriber engine uses openpyxl to automatically create and append telemetry records into data/campus\_telemetry.xlsx.

- Automatic Initialization: If the file does not exist, `_init_excel()` creates it with dark blue styled headers.
- 17-Column Schema: Every valid telemetry payload appends a complete row containing timestamp, device metadata, raw sensor metrics, derived comfort/air quality indices, and triggered alerts.
- Windows Concurrency & Lock Retry: On Windows, opening an Excel file creates a system lock. The `save_to_excel()` function implements a 3-attempt retry logic with a 0.5s pause to prevent data loss when the file is open.

## 6. Dashboard Design

Components of the Node-RED Dashboard (dashboard/node\_red\_flow.json):

- Temperature Gauge Widget (live data)
- Humidity Gauge Widget (live data)
- Occupancy Gauge Widget (live data)
- Alert Text Widget (latest alert for each device)
- Temperature Line Chart (historical graph for all devices)

How to import: Go to Node-RED → Click on Menu → Select “Import” → Paste content of node\_red\_flow.json → Deploy.

URL for Dashboard: <http://localhost:1880/ui>

## Dashboard Visualization

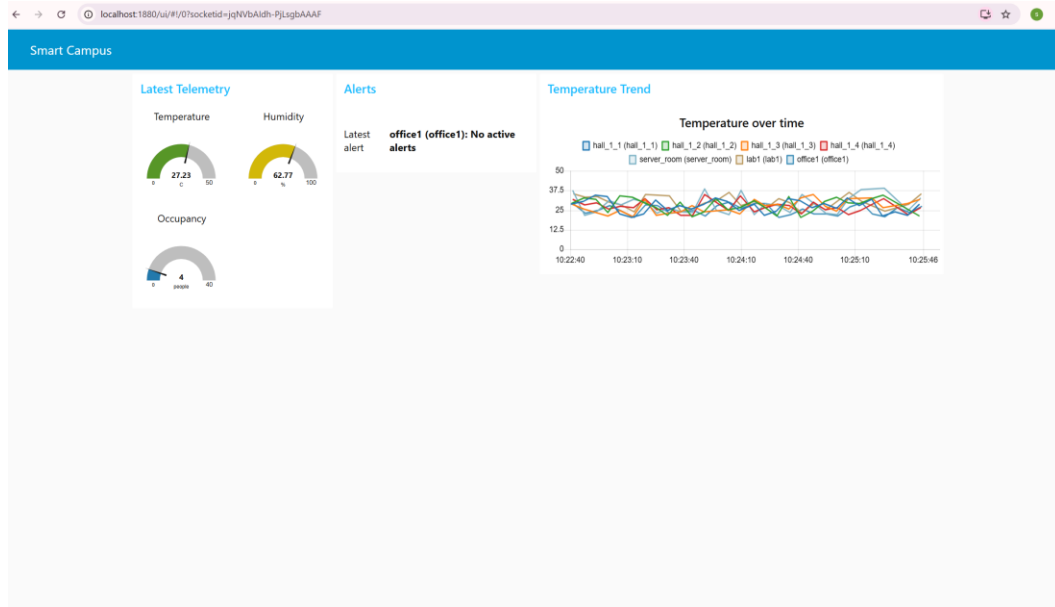


Figure 1: Smart Campus Node-RED Dashboard Live Telemetry

## 7. Security and Reliability

IoT installations experience many cyber security threat vectors arising out of the distributed architecture of sensors and the common usage of the communication channels. Typical vulnerabilities are eavesdropping (sniffing telemetry data), message spoofing (publishing fake sensor readings) and DoS attack on the broker itself. To combat above listed threats within academic scope, we design five different security mechanisms:

1. **Input Validation:** The client that subscribes to the broker checks integrity of every single JSON message – if packet contains some invalid fields and/or data items (negative occupancy rate, humidity over 100%) then packet will be silently discarded to protect databases from any corruption.
2. **Topic Naming Policy:** The client that subscribes to the broker listens only to the messages published on the strictly hierarchical pattern (campus/{building}/{room}/{deviceId}/telemetry) and ignores any attempt of writing into the system or administration topics.
3. **Log Sanitization:** Before outputting telemetry metrics to the console or logs, all string parameters are sanitized (non-alphanumeric symbols are removed from strings) in order to protect against Log Injection/RCE attacks through terminal escape sequences.

4. Configuration Isolation: Credentials of MQTT broker (username/password) are isolated in separate config.json file and not hard-coded in script for smooth transition to the secured TLS enabled private brokers.
5. Strict Topic Isolation: Using of exact device Id in publisher's topic guarantees that the failure or compromise of the device will not allow it to overwrite data of other devices.

## Reliability Features

In order to test how well the system can cope with channel impairments encountered in practice, the platform is provided with various reliability simulations of a low-power wireless network as follows:

- Packet Loss: Represents a model of the Rayleigh fading channel or path outage by random loss of 5% of the telemetry packets that have been published. The purpose is to check whether both the subscriber and dashboard can keep their states consistent without crashes.
- Propagation and Queuing Latency: Adds an arbitrary amount of latency (between 50ms to 350ms) to publish the packet in order to simulate the effects of network congestion and CSMA/CA backoffs in the Zigbee network environment.
- Energy and Battery Expenditure: Models the energy expenditure per packet transmission. If the battery life of the device goes below 20%, then the state of the device becomes LOW\_BATTERY.
- Network Failure and Disconnection: The subscriber has special listeners (on\_disconnect callbacks) to detect any network failure and notify immediately through logging the broker offline event.

## 8. Testing and Results

| Test Case                         | Description                           | Expected Result                          | Status        |
|-----------------------------------|---------------------------------------|------------------------------------------|---------------|
| <b>TC-1: Normal Telemetry</b>     | Start simulator and subscriber        | [DATA] lines printed with derived values | Pass          |
| <b>TC-2: Missing Field</b>        | Publish JSON with only partial fields | [VALIDATION] Rejected message            | Pass          |
| <b>TC-3: Abnormal Temperature</b> | Publish temp=38 for server_room       | [ALERT] Server Room: High temperature    | Pass          |
| <b>TC-4: Broker Disconnection</b> | Change broker address to invalid host | Connection failure reported              | Pass (manual) |

## Execution and Test Outputs

Virtual Device Simulator Execution Output:



```
(.venv) PS S:\master\مركز دايان\ايوت\smart-campus-iot> python .\simulator\device_simulator.py
[SIM] Smart Campus simulator started. Press Ctrl+C to stop.
[MQTT] Connected to broker
[PUBLISH] [Hall 1.1] campus/floor_1/hall_1_1/hall_1_1/telemetry | temp=20.09C humidity=79.19% battery=95.96%
[PUBLISH] [Hall 1.2] campus/floor_1/hall_1_2/hall_1_2/telemetry | temp=31.72C humidity=85.36% battery=93.96%
[PUBLISH] [Hall 1.3] campus/floor_1/hall_1_3/hall_1_3/telemetry | temp=27.44C humidity=45.14% battery=90.96%
[PUBLISH] [Hall 1.4] campus/floor_1/hall_1_4/hall_1_4/telemetry | temp=33.77C humidity=77.63% battery=88.96%
[PUBLISH] [Server Room] campus/it_building/server_room/server_room/telemetry | temp=34.62C humidity=52.7% battery=98.96%
[PUBLISH] [Lab 1] campus/engineering_building/lab1/lab1/telemetry | temp=23.66C humidity=64.65% battery=91.96%
[PUBLISH] [Office 1] campus/admin_building/office1/office1/telemetry | temp=20.62C humidity=68.53% battery=87.96%
[PUBLISH] [Hall 1.1] campus/floor_1/hall_1_1/hall_1_1/telemetry | temp=32.66C humidity=51.0% battery=95.92%
[PUBLISH] [Hall 1.2] campus/floor_1/hall_1_2/hall_1_2/telemetry | temp=22.79C humidity=73.86% battery=93.92%
[PUBLISH] [Hall 1.3] campus/floor_1/hall_1_3/hall_1_3/telemetry | temp=34.64C humidity=73.47% battery=90.92%
[PUBLISH] [Hall 1.4] campus/floor_1/hall_1_4/hall_1_4/telemetry | temp=30.54C humidity=43.71% battery=88.92%
[PUBLISH] [Server Room] campus/it_building/server_room/server_room/telemetry | temp=32.72C humidity=79.11% battery=98.92%
[PUBLISH] [Lab 1] campus/engineering_building/lab1/lab1/telemetry | temp=35.67C humidity=57.81% battery=91.92%
[ZIGBEE] Simulated packet loss for office1
[PUBLISH] [Hall 1.1] campus/floor_1/hall_1_1/hall_1_1/telemetry | temp=23.14C humidity=37.02% battery=95.88%
[PUBLISH] [Hall 1.2] campus/floor_1/hall_1_2/hall_1_2/telemetry | temp=30.09C humidity=40.56% battery=93.88%
[PUBLISH] [Hall 1.3] campus/floor_1/hall_1_3/hall_1_3/telemetry | temp=30.95C humidity=37.53% battery=90.88%
[PUBLISH] [Hall 1.4] campus/floor_1/hall_1_4/hall_1_4/telemetry | temp=22.38C humidity=57.5% battery=88.88%
[PUBLISH] [Server Room] campus/it_building/server_room/server_room/telemetry | temp=32.87C humidity=83.52% battery=98.88%
[PUBLISH] [Lab 1] campus/engineering_building/lab1/lab1/telemetry | temp=35.94C humidity=81.79% battery=91.88%
[PUBLISH] [Office 1] campus/admin_building/office1/office1/telemetry | temp=23.77C humidity=68.25% battery=87.92%
[PUBLISH] [Hall 1.1] campus/floor_1/hall_1_1/hall_1_1/telemetry | temp=22.5C humidity=39.64% battery=95.84%
[PUBLISH] [Hall 1.2] campus/floor_1/hall_1_2/hall_1_2/telemetry | temp=28.42C humidity=61.67% battery=93.84%
```

Figure 2: Python Virtual Device Simulator Console

## MQTT Subscriber and Analytics Processor Output:

```
PS S:\master\مركز دايان\ايوت\smart-campus-iot> .venv\Scripts\activate
(.venv) PS S:\master\مركز دايان\ايوت\smart-campus-iot> python .\processing\subscriber.py
[PROCESSOR] Starting subscriber...
[MQTT] Connected. Subscribing to campus/+/+/telemetry
[DATA] Hall 1.1 | temp=20.09C humidity=79.19% occupancy=35 comfort_index=24.05 air_quality=MODERATE
[DATA] Hall 1.2 | temp=31.72C humidity=85.36% occupancy=30 comfort_index=35.99 air_quality=GOOD
[ALERT] Hall 1.2: Abnormal humidity
[DATA] Hall 1.3 | temp=27.44C humidity=45.14% occupancy=37 comfort_index=29.7 air_quality=GOOD
[DATA] Hall 1.4 | temp=33.77C humidity=77.63% occupancy=30 comfort_index=37.65 air_quality=MODERATE
[DATA] Server Room | temp=34.62C humidity=52.7% occupancy=1 comfort_index=37.25 air_quality=GOOD
[DATA] Lab 1 | temp=23.66C humidity=64.65% occupancy=6 comfort_index=26.89 air_quality=MODERATE
[DATA] Office 1 | temp=20.62C humidity=68.53% occupancy=5 comfort_index=24.05 air_quality=MODERATE
[DATA] Hall 1.1 | temp=32.66C humidity=51.0% occupancy=28 comfort_index=35.21 air_quality=MODERATE
[DATA] Hall 1.2 | temp=22.79C humidity=73.86% occupancy=9 comfort_index=26.48 air_quality=MODERATE
[DATA] Hall 1.3 | temp=34.64C humidity=73.47% occupancy=11 comfort_index=38.31 air_quality=POOR
[ALERT] Hall 1.3: Poor air quality
[DATA] Hall 1.4 | temp=30.54C humidity=43.71% occupancy=8 comfort_index=32.73 air_quality=GOOD
[DATA] Server Room | temp=32.72C humidity=79.11% occupancy=1 comfort_index=36.68 air_quality=MODERATE
[DATA] Lab 1 | temp=35.67C humidity=57.81% occupancy=3 comfort_index=38.56 air_quality=GOOD
[ALERT] Lab 1: High temperature
[DATA] Hall 1.1 | temp=23.14C humidity=37.02% occupancy=39 comfort_index=24.99 air_quality=MODERATE
[DATA] Hall 1.2 | temp=30.09C humidity=40.56% occupancy=33 comfort_index=32.12 air_quality=GOOD
[DATA] Hall 1.3 | temp=30.95C humidity=37.53% occupancy=0 comfort_index=32.83 air_quality=GOOD
[DATA] Hall 1.4 | temp=22.38C humidity=57.5% occupancy=13 comfort_index=25.25 air_quality=MODERATE
[DATA] Server Room | temp=32.87C humidity=83.52% occupancy=0 comfort_index=37.05 air_quality=MODERATE
[ALERT] Server Room: Empty room with lights on
[ALERT] Server Room: Abnormal humidity
[DATA] Lab 1 | temp=35.94C humidity=81.79% occupancy=1 comfort_index=40.03 air_quality=MODERATE
[ALERT] Lab 1: High temperature
[ALERT] Lab 1: Abnormal humidity
```

Figure 3: Python MQTT Subscriber & Alerts Console

| recorded_at         | deviceid    | label       | building             | room        | temperature_C | humidity_% | occupancy | light_level | air_quality_raw | battery_% | status         | comfort_index | occupancy_status | air_quality_status    | abnormal_temp |
|---------------------|-------------|-------------|----------------------|-------------|---------------|------------|-----------|-------------|-----------------|-----------|----------------|---------------|------------------|-----------------------|---------------|
| 2026-07-22 23:34:38 | hall_1_2    | Hall 1.2    | floor_1              | hall_1_2    | 22.14         | 49.54      | 17        | 32.96       | 54.52           | 93.96 OK  | 24.62 OCCUPIED | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:34:38 | hall_1_3    | Hall 1.3    | floor_1              | hall_1_3    | 25.11         | 60.68      | 21        | 10.11       | 85.88           | 90.96 OK  | 28.14 OCCUPIED | MODERATE      | FALSE            | None                  |               |
| 2026-07-22 23:34:38 | hall_1_4    | Hall 1.4    | floor_1              | hall_1_4    | 20.4          | 85.97      | 10        | 49.96       | 112.28          | 88.96 OK  | 24.7 OCCUPIED  | MODERATE      | FALSE            | Abnormal humidity     |               |
| 2026-07-22 23:34:38 | lab1        | Lab 1       | engineering_building | lab1        | 23.39         | 56.78      | 7         | 70.42       | 121.09          | 91.96 OK  | 26.23 OCCUPIED | POOR          | FALSE            | Poor air quality      |               |
| 2026-07-22 23:34:38 | hall_1_1    | Hall 1.1    | floor_1              | hall_1_1    | 27.57         | 70.66      | 10        | 74.29       | 72.12           | 95.96 OK  | 31.1 OCCUPIED  | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:34:38 | server_room | Server Room | it_building          | server_room | 31.83         | 84.68      | 0         | 58.04       | 133.07          | 98.96 OK  | 36.06 EMPTY    | POOR          | FALSE            | Poor air quality, Abn |               |
| 2026-07-22 23:34:38 | office1     | Office 1    | admin_building       | office1     | 22.56         | 68.94      | 6         | 72.92       | 62.67           | 87.96 OK  | 26.01 OCCUPIED | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:34:38 | hall_1_1    | Hall 1.1    | floor_1              | hall_1_1    | 21.75         | 60.41      | 1         | 66.67       | 35.88           | 95.92 OK  | 24.77 OCCUPIED | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:34:38 | hall_1_2    | Hall 1.2    | floor_1              | hall_1_2    | 22.73         | 36.89      | 34        | 65.82       | 122.68          | 93.92 OK  | 24.57 OCCUPIED | POOR          | FALSE            | Poor air quality      |               |
| 2026-07-22 23:34:39 | hall_1_3    | Hall 1.3    | floor_1              | hall_1_3    | 31.41         | 42.07      | 14        | 19.41       | 37.29           | 90.92 OK  | 33.51 OCCUPIED | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:34:39 | hall_1_4    | Hall 1.4    | floor_1              | hall_1_4    | 28.86         | 71.81      | 27        | 13.97       | 104.55          | 88.92 OK  | 32.45 OCCUPIED | MODERATE      | FALSE            | None                  |               |
| 2026-07-22 23:34:39 | server_room | Server Room | it_building          | server_room | 27.07         | 58.42      | 0         | 23.29       | 106.95          | 98.92 OK  | 29.99 EMPTY    | MODERATE      | FALSE            | None                  |               |
| 2026-07-22 23:34:40 | lab1        | Lab 1       | engineering_building | lab1        | 25.33         | 42.05      | 3         | 36.04       | 123.75          | 91.92 OK  | 27.43 OCCUPIED | POOR          | FALSE            | Poor air quality      |               |
| 2026-07-22 23:34:40 | office1     | Office 1    | admin_building       | office1     | 25.69         | 53.07      | 6         | 57.31       | 38.62           | 87.92 OK  | 28.34 OCCUPIED | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:34:45 | hall_1_1    | Hall 1.1    | floor_1              | hall_1_1    | 20.44         | 45.49      | 38        | 98.3        | 89.14           | 95.88 OK  | 22.71 OCCUPIED | MODERATE      | FALSE            | None                  |               |
| 2026-07-22 23:34:45 | hall_1_2    | Hall 1.2    | floor_1              | hall_1_2    | 34.8          | 55.67      | 29        | 51.34       | 112.86          | 93.88 OK  | 37.58 OCCUPIED | MODERATE      | FALSE            | None                  |               |
| 2026-07-22 23:34:46 | hall_1_3    | Hall 1.3    | floor_1              | hall_1_3    | 26.65         | 37.8       | 21        | 44.9        | 107.33          | 90.88 OK  | 28.54 OCCUPIED | MODERATE      | FALSE            | None                  |               |
| 2026-07-22 23:34:46 | hall_1_4    | Hall 1.4    | floor_1              | hall_1_4    | 31.73         | 79.18      | 25        | 99.74       | 120.64          | 88.88 OK  | 35.69 OCCUPIED | POOR          | FALSE            | Poor air quality      |               |
| 2026-07-22 23:34:52 | hall_1_1    | Hall 1.1    | floor_1              | hall_1_1    | 23.18         | 57.59      | 0         | 74.84       | 61.16           | 95.84 OK  | 25.06 EMPTY    | GOOD          | FALSE            | Empty room with lig   |               |
| 2026-07-22 23:34:52 | hall_1_2    | Hall 1.2    | floor_1              | hall_1_2    | 32.47         | 84.14      | 9         | 30.85       | 124.05          | 93.84 OK  | 36.68 OCCUPIED | POOR          | FALSE            | Poor air quality, Abn |               |
| 2026-07-22 23:34:52 | hall_1_3    | Hall 1.3    | floor_1              | hall_1_3    | 34.94         | 35.62      | 35        | 98.02       | 49.79           | 90.84 OK  | 36.72 OCCUPIED | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:34:52 | hall_1_4    | Hall 1.4    | floor_1              | hall_1_4    | 23.43         | 70.98      | 6         | 86.99       | 58.23           | 88.84 OK  | 26.98 OCCUPIED | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:34:53 | server_room | Server Room | it_building          | server_room | 31.59         | 38.35      | 0         | 66.81       | 130.66          | 98.84 OK  | 33.51 EMPTY    | POOR          | FALSE            | Empty room with lig   |               |
| 2026-07-22 23:34:53 | lab1        | Lab 1       | engineering_building | lab1        | 28.29         | 37.65      | 11        | 64.21       | 147.08          | 91.84 OK  | 30.17 OCCUPIED | POOR          | FALSE            | Poor air quality      |               |
| 2026-07-22 23:34:58 | hall_1_1    | Hall 1.1    | floor_1              | hall_1_1    | 23.8          | 81.6       | 27        | 57.76       | 61.74           | 95.8 OK   | 27.88 OCCUPIED | GOOD          | FALSE            | Abnormal humidity     |               |
| 2026-07-22 23:34:58 | hall_1_2    | Hall 1.2    | floor_1              | hall_1_2    | 20.43         | 62.51      | 32        | 15.28       | 104.41          | 93.8 OK   | 23.56 OCCUPIED | MODERATE      | FALSE            | None                  |               |
| 2026-07-22 23:34:58 | hall_1_3    | Hall 1.3    | floor_1              | hall_1_3    | 27.14         | 74.66      | 14        | 99.53       | 37.26           | 90.8 OK   | 30.87 OCCUPIED | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:35:06 | lab1        | Lab 1       | engineering_building | lab1        | 26.49         | 58.58      | 9         | 81.95       | 125.24          | 91.76 OK  | 29.42 OCCUPIED | POOR          | FALSE            | Poor air quality      |               |
| 2026-07-22 23:35:06 | office1     | Office 1    | admin_building       | office1     | 27.15         | 41.91      | 1         | 31.08       | 91.1            | 87.76 OK  | 29.25 OCCUPIED | MODERATE      | FALSE            | None                  |               |
| 2026-07-22 23:35:12 | hall_1_1    | Hall 1.1    | floor_1              | hall_1_1    | 32.25         | 40         | 26        | 34.15       | 35.5            | 95.72 OK  | 34.25 OCCUPIED | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:35:18 | hall_1_1    | Hall 1.1    | floor_1              | hall_1_1    | 21.09         | 41.07      | 13        | 23.37       | 35.21           | 95.68 OK  | 23.14 OCCUPIED | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:35:18 | hall_1_2    | Hall 1.2    | floor_1              | hall_1_2    | 30.63         | 71.95      | 35        | 73.28       | 73.1            | 93.68 OK  | 34.23 OCCUPIED | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:35:18 | hall_1_4    | Hall 1.4    | floor_1              | hall_1_4    | 29.38         | 46.56      | 32        | 69.03       | 113.22          | 88.68 OK  | 31.71 OCCUPIED | MODERATE      | FALSE            | None                  |               |
| 2026-07-22 23:35:31 | hall_1_1    | Hall 1.1    | floor_1              | hall_1_1    | 24.52         | 77.27      | 8         | 88.03       | 83.4            | 95.6 OK   | 28.38 OCCUPIED | MODERATE      | FALSE            | None                  |               |
| 2026-07-22 23:35:31 | hall_1_2    | Hall 1.2    | floor_1              | hall_1_2    | 28.44         | 62.84      | 28        | 82.09       | 49.1            | 93.6 OK   | 31.53 OCCUPIED | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:35:31 | hall_1_3    | Hall 1.3    | floor_1              | hall_1_3    | 34.14         | 35.63      | 40        | 48.75       | 119.66          | 90.64 OK  | 35.92 OCCUPIED | MODERATE      | FALSE            | None                  |               |
| 2026-07-22 23:35:31 | hall_1_4    | Hall 1.4    | floor_1              | hall_1_4    | 21.66         | 80.07      | 7         | 41.15       | 99.49           | 88.6 OK   | 25.66 OCCUPIED | MODERATE      | FALSE            | Abnormal humidity     |               |
| 2026-07-22 23:35:32 | lab1        | Lab 1       | engineering_building | lab1        | 32.53         | 47.37      | 4         | 59.72       | 70.65           | 91.6 OK   | 34.9 OCCUPIED  | GOOD          | FALSE            | None                  |               |
| 2026-07-22 23:35:32 | server_room | Server Room | it_building          | server_room | 27.06         | 70.48      | 0         | 60.33       | 119.28          | 98.6 OK   | 30.58 EMPTY    | MODERATE      | FALSE            | Empty room with lig   |               |
| 2026-07-22 23:35:32 | office1     | Office 1    | admin_building       | office1     | 25.44         | 39.93      | 1         | 71.45       | 95.35           | 87.6 OK   | 27.44 OCCUPIED | MODERATE      | FALSE            | None                  |               |
| 2026-07-22 23:35:37 | hall_1_2    | Hall 1.2    | floor_1              | hall_1_2    | 21.36         | 59.19      | 30        | 28.32       | 82.82           | 93.56 OK  | 24.32 OCCUPIED | MODERATE      | FALSE            | None                  |               |

Figure 4: Excel Persistent Data Storage

## 9. Challenges and Limitations

The biggest issue is how one can make sure that there is a balance between realism and simplicity. Indeed, the real world Zigbee network has to be modeled on routing, interference, mesh architecture, as well as device association. In this work, however, we only focus on those features which will provide us with necessary educational insight: packet loss, delay, RSSI, link quality, energy, and battery drainage.

Another limitation is related to the usage of HiveMQ broker, which is public. The system in question should have its own private broker with SSL, authentication, and authorization.

## 10. Conclusion and Future Work

The project is a full implementation of an IoT workflow using only virtual devices in Python for 7 campus sites. MQTT protocol is used for lightweight messaging. The subscriber verifies the data, processes the analytics and issues alerts. The Node-RED Dashboard enables real-time data visualization.

Suggestions for future development:

- Implement storage of telemetry in SQLite or InfluxDB for analytics purposes.
- Implement TLS and authentication of MQTT clients.

- Design a custom web dashboard with real-time visualizations.
- Introduce device registration process and access management.
- Develop predictive analytics for temperature and air quality.

## 11. Reflection

This project was a valuable practical learning experience and it helped me link theoretical IoT concepts with real implementation. Before this I had a high-level idea of IoT but I had never implemented a full system end-to-end from sensor data generation to dashboard visualization.

During the development of the Smart Campus platform, I learned to work directly with the MQTT protocol, especially its publish/subscribe model, which is quite different from the typical request-response HTTP communication. I also learnt how to create a structured topic hierarchy to organize device data in a logical and scalable manner. Understanding the way in which JSON payloads are validated and processed at the edge layer was an eye-opener for me regarding the importance of data quality in IoT systems.

One of the problems I had to solve was to get the Node-RED dashboard to parse and display the incoming MQTT messages correctly, in real time. Another challenge was the Excel file locking issue on Windows when the file was open while the subscriber was trying to write new data.

If I were to do this project again I would use a proper time-series database like InfluxDB instead of Excel which would allow richer historical analytics and better scalability. I would also configure a private MQTT broker with TLS and authentication instead of the public HiveMQ broker.

This project helped me understand IoT architecture layers, wireless communication tradeoffs, and practical challenges of building reliable data pipelines. It also reminded me how simulation-based development can efficiently validate system design before physical hardware deployment.

## 12. AI Usage Disclosure and Student Reflection

### Academic AI Disclosure

We transparently disclose the use of AI assistance (Antigravity AI, powered by Google DeepMind) during the development of this project.

| Project Component             | AI Contribution Role                            | Estimated AI %        |
|-------------------------------|-------------------------------------------------|-----------------------|
| simulator/device_simulator.py | Code generation of VirtualDevice class & Zigbee | ~75% AI / 25% Student |

## model

|                                  |                                                   |                       |
|----------------------------------|---------------------------------------------------|-----------------------|
| processing/subscriber.py         | Validation rules, openpyxl logic, alert handling  | ~80% AI / 20% Student |
| simulator/config.json            | JSON schema design and sensor range configuration | ~70% AI / 30% Student |
| dashboard/node_red_flow.json     | Node-RED flow setup and dashboard layout          | ~20% AI / 80% Student |
| Testing, Running, & Verification | Executing code, Node-RED deployment, manual tests | 0% AI / 100% Student  |

## 13. References

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2. Eclipse Paho. Paho MQTT Python Client. <https://eclipse.dev/paho/>
3. Node-RED Documentation. <https://nodered.org/docs/>
4. HiveMQ Public MQTT Broker. <https://www.hivemq.com/mqtt/public-mqtt-broker/>
5. IEEE 802.15.4 Standard Overview. <https://standards.ieee.org/ieee/802.15.4/11041/>
6. Connectivity Standards Alliance Zigbee. <https://csa-iot.org/all-solutions/zigbee/>